

Ques 2: a. The condition is that $d < \frac{1}{3} N^{\frac{1}{4}}$

b. $N = 221503$, $e = 132317$

$$\frac{e}{N} = \frac{132317}{221503} \approx 0.59735... = 0 + \epsilon_0 \quad \text{The convergent is } \frac{A}{B} = 0$$

$$\frac{1}{\epsilon_0} = \frac{1}{0.59735} \approx 1.67406... = 1 + \epsilon_1 \quad \text{The convergent is } \frac{A}{B} = 0 + \frac{1}{1} = 1$$

$$C = \frac{eB - 1}{A} = \frac{132317 \cdot 1 - 1}{1} = 132316$$

$$\begin{aligned} P(x) &= x^2 - (N - C + 1)x + N \\ &= x^2 - (221503 - 132316 + 1)x + 221503 \\ &= x^2 - 89188x + 221503, \text{ which has no integer roots.} \end{aligned}$$

$$\frac{1}{\epsilon_1} = \frac{1}{0.67406} \approx 1.48354... = 1 + \epsilon_2 \quad \text{The convergent is } \frac{A}{B} = \frac{1}{1 + \frac{1}{2}} = \frac{1}{2}$$

$$C = \frac{eB - 1}{A} = \frac{132317 \cdot 2 - 1}{1} = 264633$$

$$\begin{aligned} P(x) &= x^2 - (221503 - 264633 + 1)x + 221503 \\ &= x^2 + 43129x + 221503 \end{aligned}$$

$$\frac{1}{\epsilon_2} = \frac{1}{0.48354} \approx 2.068081... = 2 + \epsilon_3 \quad \text{The convergent is } \frac{A}{B} = 0 + \frac{1}{1 + \frac{1}{2}} = \frac{2}{3}$$

$$C = \frac{eB-1}{A} = \frac{132317.5-1}{3} = 220528$$

$$P(x) = x^2 - (221503 - 220528 + 1)x + 221503$$

$$= x^2 - 976x + 221503$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a=1, b=-976, c=221503$$

$$x = \frac{-(-976) \pm \sqrt{(-976)^2 - 4(1)(221503)}}{2(1)}$$

$$= 976 \pm 258$$

2

$$\frac{976+258}{2}$$

$$\frac{976-258}{2}$$

2

2

$$= 617$$

$$= 359$$

$$p=617, q=359, d=5$$

$$P(x) = (x-617)(x-359)$$

3. $N = 1271$, $x_0 = y_0 = 3$, $P(x) = x^2 + 4$

$$x_1 = 13 \bmod 1271 = 13, \quad y_1 = 173 \bmod 1271 = 173$$

$$\gcd(1x_1 - y_1, N) = \gcd(160, 1271) = 1$$

$$x_2 = 173 \bmod 1271 = 173, \quad y_2 = 895984493 \bmod 1271 = 669$$

$$\gcd(1x_2 - y_2, N) = \gcd(496, 1271) = 31$$

First factor is 31, second factor is 41
 $N = 31 \times 41 = 1271$

4. $p = 101$, $g = 2$, $h = g^x \bmod p = 27$, $x = 7$

$$C = [c_1, c_2] = [100, 70]$$

$$m = c_2 \cdot c_1^{-x} \pmod{p}$$

$$m = 70 \cdot 100^{-7} \pmod{101}$$

$$70 \cdot 100^{-7} = x \pmod{101}$$

$$\text{Calculate } 100^7 \bmod 101 = 100$$

$$\text{Modular inverse of } 100^7 \bmod 101$$

$$= 100^{-1} \equiv 100 \pmod{101}$$

$$70 \cdot 100^{-7} \equiv 70 \cdot 100 \pmod{101}$$

$$= 31$$

$$x \equiv 31 \pmod{101}$$

plaintext was 31.

1. $p=23$, $q=37$, $N=pq=851$, $e=293$

0. $a = (p-1)(q-1)$
 $= (22)(36)$
 $= 792$

i	Quotient $q = r_1 \div r_2$	Remainder $r = r_1 - q \cdot r_2$	$s = s_1 \cdot q - s_2$	$t = t_1 - q \cdot t_2$
1		293	1	0
2		792	0	1
3	0	293	1	0
4	2	206	-2	1
5	1	87	3	-1
6	2	32	-8	3
7	2	23	19	-7
8	1	9	-27	10
9	2	5	73	-27
10	1	4	-100	36
→ 11	1	1	173	-64
12	4	0	-792	293

This tells us that $d = \gcd(a, b) = 1$ and $-64a + 173b = 1$

Hence $173e = 1 + 64\phi(N)$ and $d=173$

b. $e=293$

$$293 = 100100101_2$$

$$e = 1 \times 2^8 + 0 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 2^8 + 2^5 + 2^2 + 2^0$$

c. $d = 173$

$$173 = 10101101_2$$

$$d = 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 2^7 + 2^5 + 2^3 + 2^2 + 2^0$$

d. $c(m) = m^e \bmod n$

$$= (77)^{203} \bmod 851$$

$$m \cdot m^2 \cdot m^{2^5} \cdot m^{2^8} \bmod N$$

$$m^2 \bmod N = (77)^2 \bmod N = 823$$

$$m^{2^2} \bmod N = (77)^{2^2} \bmod N = 784 \checkmark$$

$$m^{2^3} \bmod N = (77)^{2^3} \bmod N = 234$$

$$m^{2^4} \bmod N = (77)^{2^4} \bmod N = 292$$

$$m^{2^5} \bmod N = (77)^{2^5} \bmod N = 164 \checkmark$$

$$m^{2^6} \bmod N = (77)^{2^6} \bmod N = 515$$

$$m^{2^7} \bmod N = (77)^{2^7} \bmod N = 564$$

$$m^{2^8} \bmod N = (77)^{2^8} \bmod N = 673 \checkmark$$

$$(77)(784)(164)(673) \bmod 851 = 58 \text{ is the encryption of } m$$

e. $m(c) = c^d \bmod N$

$$= (286)^{173} \bmod 851$$

$$c \cdot c^{2^2} \cdot c^{2^3} \cdot c^{2^5} \cdot c^{2^7} \bmod N = (286)(639)(692)(729)(285) \bmod 851$$

= 159 is the
decryption of c

$$c^2 \bmod N = 100 \quad c^{2^6} \bmod N = 417$$

$$c^{2^2} \bmod N = 639 \checkmark \quad c^{2^7} \bmod N = 285 \checkmark$$

$$c^{2^3} \bmod N = 692 \checkmark \quad c^{2^8} \bmod N = 360$$

$$c^{2^4} \bmod N = 602$$

$$c^{2^5} \bmod N = 729 \checkmark$$